



SEMI E116-0705

PROVISIONAL SPECIFICATION FOR EQUIPMENT PERFORMANCE TRACKING

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1 Purpose

1.1 This document provides specification for implementing basic equipment performance tracking for production equipment. Provisions in this document enable the host computer to track basic equipment performance in an automated and consistent manner, without operator or host input. This document provides specification for equipment suppliers to:

- Track basic equipment states (no operator or host input required).
- Track basic equipment states in a modular manner, for both major modules and the overall equipment.
- Report basic equipment state changes to a host computer, at both module and equipment level.
- Report equipment's time in state to a host computer, at both module and equipment level.
- Report reasons to a host computer for why equipment is blocked from performing its task, at both module and equipment level.

1.2 Equipment users require the ability to track equipment performance without dependence on user input to eliminate inaccuracies due to incorrect or untimely input from the user. The ability to track equipment performance without user input is essential in 300 mm wafer factories where operational scenarios require minimal manual interaction. EPT defines concepts, behavior, and message services that enable the host computer to obtain the equipment data required for equipment performance tracking in an automated and consistent manner without operator or host input.

1.3 EPT enables factory managers to identify the current states of factory equipment, both at the equipment level and at the module level (e.g., processing chambers), without dependence on user input. EPT enables factory engineers to evaluate the time that equipment and modules spent in different states and identify areas for improvement. EPT enables factory engineers to obtain directly from the equipment the reasons why the equipment or module is prevented from performing. EPT provides industrial engineers the equipment data which, when combined with external data from the Manufacturing Execution System (MES), will enable accurate calculation of SEMI E10 states and SEMI E79 metrics at the equipment-level and the module-level. EPT enables automation engineers to develop reusable host interfaces by using a standardized collection event and data variables to collect equipment state data.

2 Scope

2.1 The Scope of this standard is to define equipment behavior states and the data required to track basic equipment performance for production equipment. These requirements are intended to facilitate equipment-level and module-level state tracking and to communicate state information to the host for simple equipment performance tracking, without requiring host or operator input.

2.2 Specifically, this document provides:

- An Equipment Performance Tracking (EPT) state model that defines triggers for state changes.
- Specification of data variables required to communicate basic equipment performance data to the host computer.
- Specification of event messages used to communicate basic equipment performance data to the host computer.
- Requirements for EPT compliance.



2.3 This standard specifies the concepts, behavior, and message services that enable the host computer to obtain the equipment data required for equipment performance tracking in an automated and consistent manner, without operator or host input. It does not specify the report of SEMI E10 states from the equipment to a host computer, as this information requires user input and is already specified by SEMI E58.

2.4 This standard does not conflict with SEMI E58 nor does it inhibit the equipment's ability to report out SEMI E10 states via SEMI E58.

2.5 This standard is a building block towards SEMI E10 and SEMI E79 by providing accurate equipment information required for SEMI E10 and SEMI E79 metrics, without dependence on operator or host input, eliminating inaccuracies due to incorrect or untimely input from the user. EPT assists both SEMI E10 and SEMI E79 by providing a modular approach to equipment performance tracking, allowing the state of the equipment to be determined by the states of its major modules. EPT assists SEMI E79 by providing task-level detail of the equipment or module's current activity, allowing performance metrics to be tracked at the task level.

2.6 This is a provisional standard. In order to have the provisional status removed, the following areas must be completed:

- Equipment Interlock Information
- Equipment Process Monitoring Information
- Equipment Maintenance Trigger Information
- Application of EPT State Model to load ports and internal buffers

NOTICE: This standard does not purport to address safety issues, if any, associated with its use. It is the responsibility of the users of this standard to establish appropriate safety and health practices and determine the applicability of regulatory or other limitations prior to use.

3 Limitations

3.1 This standard applies to semiconductor production equipment. Other types of equipment have not been examined. However, it may be used for other types of equipment when applicable.

4 Referenced Standards and Documents

4.1 SEMI Standards

SEMI E10 — Specification for Definition and Measurement of Equipment Reliability, Availability, and Maintainability (RAM)

SEMI E39 — Object Services Standard: Concepts, Behavior, and Services

SEMI E30 — Generic Model for Communications and Control of Manufacturing Equipment (GEM)

SEMI E58 — Automated Reliability, Availability, and Maintainability Standard (ARAMS): Concepts, Behavior, and Services

SEMI E79 — Standard for Definition and Measurement of Equipment Productivity

SEMI E90 — Specification for Substrate Tracking

SEMI E98 — Provisional Standard for the Object-Based Equipment Model (OBEM)

SEMI 101 — Provisional Guide for EFEM Functional Structure Model

NOTICE: Unless otherwise indicated, all documents cited shall be the latest published versions.

5 Terminology

5.1 Abbreviations and Acronyms

5.1.1 *EFEM* — Equipment Front End Module

5.1.2 *EPT* — Equipment Performance Tracking

5.1.3 *GEM* — Generic Equipment Model [SEMI E30]

5.2 Definitions

5.2.1 *EPT module* — a major component of the equipment that affects processing or throughput.

5.2.1.1 For purposes of simplification, an EPT module executes one and only one task at a time. Each EPT Module has an EPT state model that is maintained by the equipment.

5.2.2 *EPT state* — the state of IDLE, BUSY, or BLOCKED within EPT state model.

5.2.3 *event report* — a message the equipment sends to the host on the occurrence of a collection event.

5.2.4 *fault* — an exception.

5.2.5 *host* — the factory computer system or an intermediate system that represents the factory and the operator to the equipment.

5.2.6 *intended function* — a manufacturing function that the equipment was built to perform. This includes transport functions for transport equipment, measurement functions for metrology equipment, as well as process functions such as physical vapor deposition and wire bonding. Complete equipment may have more than one intended function. [SEMI E58]

5.2.7 *material* — the basic unit of process. For the purposes of this standard, material is a set of one or more substrates.

5.2.8 *module* — a major component of equipment that contains at least one material location and performs some task on material. Equipment modules may be aggregates of equipment subsystems, I/O devices, and other modules. [SEMI E98]

5.2.9 *substrate* — the basic unit of material on which work is performed to create a product. Examples include wafers, die, plates used for masks, flat panels, circuit boards, and leadframes. [SEMI E90]

5.2.10 *task* — a planned and repeatable activity with an expected duration and a definite beginning and end (e.g., Move wafer from cassette to stage, Pre-align wafer, Align reticle, Preheat chamber, Increase vacuum). NOTE: Actual durations may vary.

5.2.11 *trigger* — an event that causes a change in the state of the equipment. Examples are changes in sensor readings, alarms, and messages received from the host, and operator commands.

5.2.12 *unit* — any wafer, die, packaged device, or piece thereof (included product and non-product units). [SEMI E10]

5.2.13 *user* — any entity interacting with the equipment, either locally as an operator or remotely via the host. From the equipment's viewpoint, both the operator and the host represent the user. [SEMI E58]

6 Requirements

6.1 An EPT compliant implementation requires provision of certain capabilities defined by other standards: accessibility to status information, event reporting, alarm management, and provision of an internal time-and-date clock. These requirements may be satisfied through compliance to SEMI E30 for the following sets of requirements:

- Clock Services
- Event Notification
- Status Data Collection

- Equipment Constants
- Alarm Management

6.2 An EPT compliant implementation requires a documented list of all EPT modules contained within the equipment.

6.3 An EPT compliant implementation requires an EPT state model for the equipment and for each EPT module.

6.4 Each EPT module shall have a documented list of tasks that the module may perform.

7 Conventions

7.1 This document follows the conventions for state model methodology and service definitions used by the SEMI standards referenced in §4.

7.2 State Model Methodology

7.2.1 A state model has three elements: definitions of each state and sub-state, a diagram of the states showing the valid transitions between states, and a corresponding state transition table that defines the triggers for each transition. The diagram of the state model uses the Harel State Chart notation. An overview of this notation is presented in an Appendix of SEMI E30. The definition of this notation is presented in Science of Computer Programming 8, “Statecharts: A Visual Formalism for Complex Systems”, by D. Harel, 1987.

7.2.2 Transition tables are provided in conjunction with the state diagrams to explicitly describe the nature of each state transition. A transition table contains columns for Transition number, Previous State, Trigger, New State, Actions, and Comments. The “trigger” (column 3) for the transition occurs while in the “previous” state. The “actions” (column 5) includes a combination of:

- 1) Actions taken upon exit of the previous state.
- 2) Actions taken upon entry of the new state.
- 3) Actions taken that are most closely associated with the transition.

7.2.3 No differentiation is made between these cases.

#	PREVIOUS STATE	TRIGGER	NEW STATE	ACTIONS	COMMENTS
Transition #					

7.3 Variable Data Item Definition

7.3.1 The variable item dictionary contained in this document defines the variable name, description, type (format), access (read/write or read only), and comments about the variable. This is depicted in the format of the following table:

VARIABLE NAME	DESCRIPTION	TYPE	ACCESS	COMMENT

7.4 Event Description

7.4.1 The collection event description table contained in this document includes the event name, the state model transition, the description, and the data required for the event. This is depicted in the format of the following table:

EVENT NAME	STATE MODEL TRANSITION NUMBER	DESCRIPTION	REQUIRED DATA

8 Overview

8.1 This section defines the concepts that enable EPT to provide information required for determining equipment performance.

8.2 Equipment processing times, including throughput and cycle times, are determined by two factors: (1) the sequence of tasks performed by the equipment's EPT modules, including the number of each task performed, as well as the series and parallel relationships between tasks, and (2) the execution time of each individual task. By tracking equipment systems in terms of modules that perform tasks, it will be possible to precisely identify the affect of both factors toward overall system performance.

8.3 EPT enables tracking to this level of detail by introducing the concept of an EPT module and the concept of a task. EPT-compliant equipment are required to report events according to a breakdown of equipment operations into tasks and EPT modules. An EPT module is an equipment component that affects processing or throughput, and is capable of executing only one task at a time. Examples of EPT modules include process stations, process chambers, wafer handling systems, etc. A task is a planned and repeatable activity with an expected duration and a definite beginning and end. Examples of tasks include the following:

- Move wafer from cassette to stage
- Pre-align wafer
- Align reticle
- Preheat chamber
- Increase vacuum etc.

8.4 EPT-compliant equipment consist of EPT modules that perform tasks. Thus, the state of the equipment can be effectively modeled as a composition of the states of the EPT modules. In addition, EPT provides detailed information required for performance tracking by providing the task-level details for each EPT module.

8.5 EPT modules and tasks are to be defined such that each EPT module executes only one task at a time. An example is given in Related Information §R1-7. Each equipment system must have defined at least as many EPT modules as tasks that may be performed simultaneously by that system.

8.6 Each task should be defined such that the units involved (e.g., wafers, substrates, reticles), the EPT modules involved (e.g., process station, robot, etc.), and the task type (e.g., "process", "support", etc.) are constant throughout the duration of the task. A change in any of the following items should result in the start of a new task or completion of current task:

- Change in material involved (e.g., arrival of wafer, removal of wafer)
- Change of EPT modules involved (e.g., robot moves wafer, chamber processes wafer)
- Change in task type (e.g., purge recipe followed by production recipe)

NOTE 1: It is not practical for equipment to define tasks or modules at an indefinite level of detail. It is therefore recommended that in each instance for which the material involved, the EPT modules involved, and the task types are all constant, events for only one task should be reported. Any further breakdown is not required for EPT compliance. EPT places no limitations on supplemental event reporting within any task. However, these supplemental events should not be reported as EPT events.

9 EPT State Model

9.1 This section defines the basic Equipment Performance Tracking State Model, which is applied to the equipment and its EPT modules.

9.2 EPT State Model Requirements

9.2.1 The EPT state model is intended to capture the different states of the equipment and its EPT modules from an operational point of view:

- EPT module busy executing a task.
- EPT module blocked from executing a task.

- EPT module idle.

9.2.2 The equipment shall maintain an EPT state model for each major module (e.g., process stations and chambers, wafer handling systems, etc.) through which a unit passes throughout the entire equipment from removal of units from the carrier to the return of units to the carrier.

9.2.3 The equipment shall maintain an EPT state model for any major component of the equipment that impacts processing and throughput.

9.2.4 The equipment shall maintain its own EPT state model for the overall equipment.

9.2.5 The equipment is responsible for communicating all EPT state transitions.

9.2.6 The EPT state model can be applied to the Equipment Front-End Module (EFEM).

9.2.7 There are two types of EPT Modules: (1) Production EPT modules, and (2) EFEM/LoadPort EPT modules. Both types have a module-level EPT State Machine.

9.3 EPT State Model Diagram

9.3.1 Figure 1 shows the EPT state module diagram.

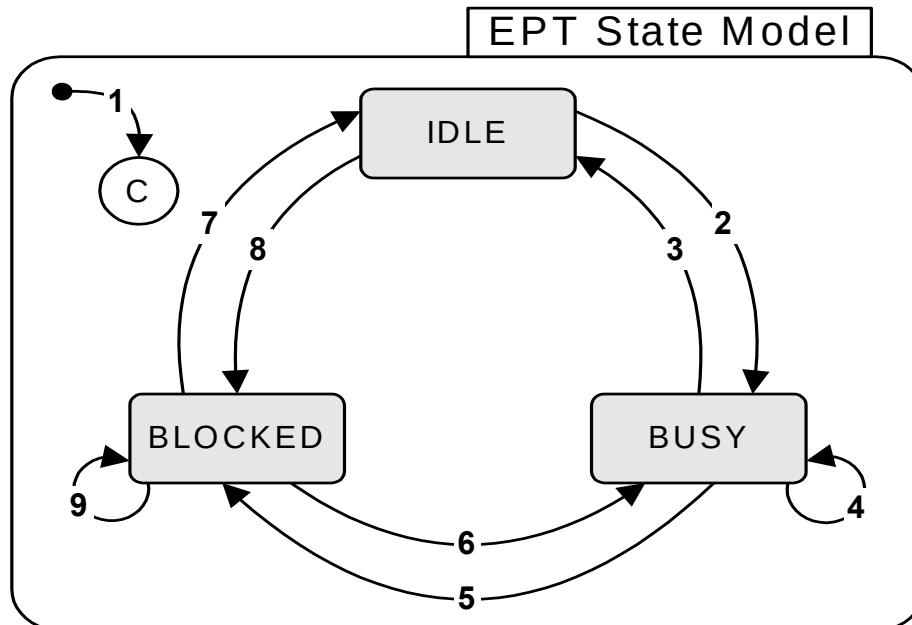


Figure 1
Equipment Performance Tracking State Model

9.4 EPT State Definitions

This section describes the EPT states.

9.4.1 IDLE

9.4.1.1 *EPT Module* — The following conditions are required for the module to be in the IDLE state:

- No material is present in the module and
- The module is not executing a task and
- No fault condition exists in the module that prevents it from starting a new task.

9.4.1.2 *Equipment* — The following conditions are required for the equipment to be in the IDLE state:

- No BLOCKED EPT modules prevent an EPT module from starting a new task, and
- There is no EPT module in the BUSY state with a task type of 1, 2, 3, 4 or 5.

9.4.1.3 For both the equipment and modules the following conditions may exist in the IDLE State:

- The equipment or module is maintaining and monitoring environmental requirements (e.g., background temperature, particle monitoring, etc).

9.4.2 BUSY

9.4.2.1 *EPT Module* — The following conditions are required for the module to be in the BUSY state:

- The module is executing a task and
- No fault condition exists in the module that prevents the execution of a task.

9.4.2.2 *Equipment* — The following condition is required for the equipment to be in the BUSY state:

- At least one EPT module is BUSY with a task type of 1, 2, 3, 4, or 5.

9.4.2.3 For both the equipment and modules the following conditions may exist in the BUSY State:

- Material is present.

9.4.3 BLOCKED

9.4.3.1 *EPT Module* — One or more of the following conditions are required for the module to be in the BLOCKED state:

- Conditions exist that do not allow the EPT module to continue or start execution of a task:
- A fault condition(s) exists that prevents the EPT module from completing its task.
- A fault condition(s) exists that prevents the EPT module from starting a new task.
- The EPT module is pausing (or aborting) as the result of a pause (or abort) directive.
- The EPT module has paused its task and is awaiting a resume directive.
- The EPT module fails to initialize upon start-up.

9.4.3.2 *Equipment* — The following condition is required for the EPT equipment to be in the BLOCKED state:

- No EPT module is BUSY with task type of 1, 2, 3, 4 or 5, and
- At least one BLOCKED EPT module prevents any EPT module from starting a new task or completing a current task.

9.4.3.3 For both the EPT equipment and modules the following conditions may exist in the BLOCKED State:

- Material is present.

9.5 EPT State Transition Table – Module Level

9.5.1 This section defines the state transitions for the EPT modules within the equipment.

9.5.2 For each status variable updated with the event change, the corresponding event variable shall also be updated (see Table 4).

Table 1 Basic EPT State Transitions – Module Level

#	PREVIOUS STATE	TRIGGER	NEW STATE	ACTIONS	COMMENTS
1	(No state)	EPT Module _i initialization completed.	IDLE [OR] BLOCKED	- No EPT event is triggered $EPTState_i = \text{IDLE}$ or BLOCKED $PreviousEPTState_i = \text{NoState}$ $PreviousTaskName_i = \text{"No Task"}$ $PreviousTaskType_i = 0$ $EPTStateTime_i = 0$	- If no Fault conditions exist and no material is present, transition to IDLE - If Fault conditions exist, or EPT module cannot be initialized, then transition to BLOCKED - If Material is present, then transition to BLOCKED
2	IDLE	EPT Module _i starts execution of a new task.	BUSY	$EPTStateChange_i$ event is triggered $EPTStateTime_i$ is calculated $EPTState_i = \text{BUSY}$ $PreviousEPTState_i = \text{IDLE}$ $TaskName_i$ is set $TaskType_i$ is set	
3	BUSY	EPT Module _i completes execution of task. [AND] Material is removed from the EPT Module _i .	IDLE	$EPTStateChange_i$ event is triggered $EPTStateTime_i$ is calculated $EPTState_i = \text{IDLE}$ $PreviousEPTState_i = \text{BUSY}$ $TaskName_i = \text{"No Task"}$ $TaskType_i = 0$ $PreviousTaskName_i$ is set $PreviousTaskType_i$ is set	
4	BUSY	EPT Module _i starts execution of a new task upon the normal completion of the previous task.	BUSY	$EPTStateChange_i$ event is triggered $TaskName_i$ is set $TaskType_i$ is set	$EPTStateTime_i$ does not change. Thus when the module transitions out of BUSY state, the value of $EPTStateTime_i$ will reflect the total time the module was BUSY. $PreviousEPTState_i$ does not change. Thus, $PreviousEPTState_i$ will continue to reflect the state of the module before it was BUSY.

#	PREVIOUS STATE	TRIGGER	NEW STATE	ACTIONS	COMMENTS
5	BUSY	A pause command is received that pauses the EPT Module_i. [OR] An abort command is received that aborts the EPT Module_i. [OR] A Fault condition occurs that prevents the EPT Module_i from executing its task.	BLOCKED	• <i>EPTStateChange_i</i> event is triggered • <i>EPTStateTime_i</i> is calculated • <i>BlockedReason_i</i> is set • <i>BlockedReasonText_i</i> is set • <i>EPTState_i</i> =BLOCKED • <i>PreviousEPTState_i</i>=BUSY	TaskName _i and TaskType _i do not change. The variables continue to reflect the information of the task that is currently blocked from executing.
6	BLOCKED	All fault conditions that prevented task execution are cleared and the EPT Module_i resumes execution of its task or starts execution of a new task. [OR] A resume command is received that enables the EPT Module_i to resume its task.	BUSY	• <i>EPTStateChange_i</i> event is triggered • <i>EPTStateTime_i</i> is calculated • <i>EPTState_i</i> = BUSY • <i>PreviousEPTState_i</i> =BLOCKED • <i>TaskName_i</i> is set • <i>TaskType_i</i> is set • <i>BlockedReason_i</i> = 0 • <i>BlockedReasonText_i</i> = “Not Blocked”	
7	BLOCKED	All fault conditions that prevented task execution are cleared. [AND] All material is removed from the EPT Module_i. [AND] EPT Module_i can begin a new task.	IDLE	• <i>EPTStateChange_i</i> event is triggered • <i>EPTStateTime_i</i> is calculated • <i>EPTState_i</i> =IDLE • <i>PreviousEPTState_i</i> =BLOCKED • <i>TaskName_i</i> = “No Task” • <i>TaskType_i</i> = 0 • <i>PreviousTaskName_i</i> is set • <i>PreviousTaskType_i</i> is set • <i>BlockedReason_i</i> = 0 • <i>BlockedReasonText_i</i> = “Not Blocked”	

#	PREVIOUS STATE	TRIGGER	NEW STATE	ACTIONS	COMMENTS
8	IDLE	Material arrives at the EPT Module _i and the EPT Module _i cannot begin executing a task on the material. [OR] A fault condition occurs which prevents the EPT Module _i from starting a new task.	BLOCKED	• <i>EPTStateChange_i</i> event is triggered • <i>EPTStateTime_i</i> is calculated • <i>BlockedReason_i</i> is set • <i>BlockedReasonText_i</i> is set • <i>EPTState_i</i> = BLOCKED • <i>PreviousEPTState_i</i> = IDLE	
9	BLOCKED	Fault conditions occur that prevent the EPT Module _i from resuming a blocked task, or starting a new task.	BLOCKED	• <i>EPTStateChange_i</i> event is triggered • <i>BlockedReason_i</i> is set • <i>BlockedReasonText_i</i> is set	<i>EPTStateTime_i</i> does not change. Thus when the module transitions out of BLOCKED state, the value of <i>EPTStateTime_i</i> will reflect the total time the module was BLOCKED from processing. <i>PreviousEPTState_i</i> does not change. Thus, <i>PreviousEPTState_i</i> will continue to reflect the state of the module before it was BLOCKED from processing.

9.6 EPT State Transition Table for Equipment

9.6.1 This section defines the State Transitions for the Equipment, in terms of its EPT modules.

Table 2 Basic EPT State Transitions — Equipment Level

#	PREVIOUS STATE	TRIGGER	NEW STATE	ACTIONS	COMMENTS
1	(No state)	Equipment initiation completed (includes initialization of EPT modules)	IDLE [OR] BLOCKED [OR] BUSY	• <i>EPTStateChange</i> event is triggered • <i>EPTState</i> = IDLE or BLOCKED or BUSY • <i>PreviousEPTState</i> = NoState • <i>EPTStateTime</i> = 0 • <i>BlockedReason</i> may be updated • <i>BlockedReasonText</i> may be updated	
2	IDLE	At least one EPT module transitions to the BUSY state with task type of 1..5.	BUSY	• <i>EPTStateChange</i> event is triggered • <i>EPTStateTime</i> is calculated • <i>EPTState</i> = BUSY • <i>PreviousEPTState</i> = IDLE	

#	PREVIOUS STATE	TRIGGER	NEW STATE	ACTIONS	COMMENTS
3	BUSY	All EPT modules that were BUSY with task type of 1..5 transition either to IDLE or to BUSY with task type other than 1..5.	IDLE	• <i>EPTStateChange</i> event is triggered • <i>EPTStateTime</i> is calculated • <i>EPTState</i> = IDLE • <i>PreviousEPTState</i> = BUSY	
4	BUSY	Any EPT module transitions to BUSY state with task type 1..5.	BUSY		
5	BUSY	{ At least one EPT module transitions to a BLOCKED state and prevents another EPT module from starting a new task or completing an existing task [AND] There are no other EPT modules BUSY with task type 1..5. } [OR] { Any EPT module that was BUSY with task type 1..5 transitions to either IDLE or BUSY with task type other than 1..5. [AND] There is at least one EPT module in BLOCKED state that is preventing another EPT module from starting a new task or completing an existing task. }	BLOCKED	• <i>EPTStateChange</i> event is triggered • <i>EPTStateTime</i> is calculated • <i>EPTState</i> = BLOCKED • <i>PreviousEPTState</i> = BUSY • <i>BlockedReason</i> is set • <i>BlockedReasonText</i> is set	
6	BLOCKED	At least one EPT module transitions to BUSY with task type of 1..5.	BUSY	• <i>EPTStateChange</i> event is triggered • <i>EPTStateTime</i> is calculated • <i>EPTState</i> = BUSY • <i>PreviousEPTState</i> = BLOCKED	

#	PREVIOUS STATE	TRIGGER	NEW STATE	ACTIONS	COMMENTS
7	BLOCKED	All BLOCKED EPT modules that prevented another EPT module from starting a new task or completing a current task no longer prevents the EPT module from doing so. [AND] No EPT module is BUSY with task type 1.. 5.	IDLE	• <i>EPTStateChange</i> event is triggered • <i>EPTStateTime</i> is calculated • <i>EPTState</i> = IDLE • <i>PreviousEPTState</i> = BLOCKED	
8	IDLE	Any EPT module transitions to BLOCKED that prevents another EPT module from starting a new task.	BLOCKED	• <i>EPTStateChange</i> event is triggered • <i>EPTStateTime</i> is calculated • <i>BlockedReason</i> is set • <i>BlockedReasonText</i> is set • <i>EPTState</i> = BLOCKED • <i>PreviousEPTState</i> = IDLE	
9	BLOCKED	Any EPT module transitions to BLOCKED that prevents or continues to prevent another EPT module from starting a new task or completing a current task.	BLOCKED	• <i>EPTStateChange</i> event is triggered • <i>EPTStateTime</i> is calculated • <i>BlockedReason</i> is set • <i>BlockedReasonText</i> is set	EPTStateTime does not change. Thus when the equipment transitions out of BLOCKED state, the value of EPTStateTime will reflect the total time the equipment was BLOCKED from processing. PreviousEPTState does not change. Thus, PreviousEPTState will continue to reflect the state of the equipment before it was BLOCKED from processing.

10 EPTTracker Objects

10.1 The equipment shall provide SEMI E39-compliant EPTTracker objects to track performance of EPT Equipment and Modules. The equipment shall provide one EPTTracker object for the equipment and one EPTTracker object for each EPT Module (both Production and EFEM/LoadPort EPT Modules). The EPTTracker object is used to maintain and update the SEMI E116 attributes each time an SEMI E116 transition occurs. These EPTTracker objects will be owned at the equipment level. These objects shall be available immediately after equipment initialization, and shall be persistent until equipment shutdown. If certain attributes are identified as persistent, the object shall restore the value of these attributes on initialization. The Host can use SEMI E39 services (e.g., GetAttr) to access EPTTracker objects. The EPTTracker object format is specified in Table 3.

Table 3 EPTTracker Object Definition

<i>Attribute Name</i>	<i>Definition</i>	<i>Access</i>	<i>Reqd</i>	<i>Form</i>
ObjType	Object type	RO	Y	Text “EPTTracker”
ObjID	Object identifier. A name assigned by the Equipment designer uniquely identifying the instance of the EPTTracker object within this Equipment.	RO	Y	Text 1 to 80 characters. Shall conform to SEMI E39 restrictions on ObjID.
BlockedReason	A numeric code that identifies the most recent blocked condition of the related EPT module.	RO	Y	Enumerated: 0 = Not Blocked 1 = Unknown 2 = Safety Threshold 3 = Error Condition 4 = Parametric Exception 5 = Aborting, Aborted 6 = Pausing, Paused 7 = Reserved 8 = Reserved 9 = Reserved
BlockedReasonText	A description of the most recent blocked condition of this Equipment or EPT module. May provide further details to BlockedReason.	RO	Y	Text 0 to 80 characters This attribute must be defined for both equipment and module EPTTracker objects. For the Equipment, a BlockedReason value is determined by the blocked module.
DisableEventOnTransition	A list of zero to nine EPT State Model transitions for which event reporting to the host are disabled.	RW	Y	List of zero to nine integers in the range 1–9.
EPTElementType	A numeric code that indicates whether this EPTTracker object refers to the equipment or an EPT module.	RO	Y	Enumerated 0 = Equipment 1 = Production EPT Module 2 = EFEM/LoadPort EPT Module
EPTState	The current EPT state of this equipment or EPT module.	RO	Y	Enumerated 0 = Idle 1 = Busy 2 = Blocked
EPTStateTime	Time spent in the previous EPT State, prior to entering the current EPTState.	RO	Y	Unsigned integer Value in seconds
EPTElementName	A human-understandable name for this equipment or module, which must be unique among all objects of EPTTracker type.	RO	Y	Text 0 to 80 characters
PreviousEPTState	The previous EPT state of this equipment or EPT module, prior to entering the current EPTState.	RO	Y	Enumerated 0 = Idle 1 = Busy 2 = Blocked 3 = No State

<i>Attribute Name</i>	<i>Definition</i>	<i>Access</i>	<i>Reqd</i>	<i>Form</i>
PreviousTaskName	Name of the EPT Task previously running on this EPT module, prior to starting the current EPT Task.	RO	Y	Text 0 to 80 characters This attribute must be defined for both equipment and module EPTTracker objects. For the Equipment, the value of the PreviousTaskName attribute may be left blank (empty string).
PreviousTaskType	Type of the EPT Task previously running on this EPT module, prior to starting the current EPT Task.	RO	Y	Enumerated: 0 = No Task 1 = Unspecified 2 = Process -- adding value (e.g., exposing) 3 = Support -- incapable of adding value (e.g., Handling/Transport) 4 = Equipment Maintenance (e.g., Equipment-initiated clean cycle) 5 = Equipment Diagnostics (e.g., Equipment-initiated health check) 6 = Waiting (e.g., chamber waiting for a robot to remove a substrate) This attribute must be defined for both equipment and module EPTTracker objects. For the Equipment, the value of the PreviousTaskType attribute may be set to 0.
TaskName	Name of the EPT Task currently running on this EPT module.	RO	Y	Text 0 to 80 characters This attribute must be defined for both equipment and module EPTTracker objects. For the Equipment, the value of the TaskName attribute may be left blank (empty string).
TaskType	The type of EPT Task currently running on this EPT module.	RO	Y	Enumerated: 0 = No Task 1 = Unspecified 2 = Process -- adding value (e.g., exposing) 3 = Support -- incapable of adding value (e.g., Handling/Transport) 4 = Equipment Maintenance (e.g., Equipment-initiated clean cycle) 5 = Equipment Diagnostics (e.g., Equipment-initiated health check) 6 = Waiting) e.g., chamber waiting for a robot to remove a substrate) This attribute must be defined for both equipment and module EPTTracker objects. For the Equipment, the value of the TaskType attribute may be set to 0.

Attribute Name	Definition	Access	Reqd	Form
TransitionTimestamp	The timestamp when the most recent transition occurred and collection event was triggered that set the tracker to its current state.	RO	Y	Formatted numeric text: timestamp format – <i>list specifics</i> 16 characters YYYYDDMMHHMMSSCC
Transition	The most recent transition that updated the tracker.	RO	Y	Enumerated: 1 = Transition 1 2 = Transition 2 ... 9 = Transition 9
TrackerEventID	Numeric identifier for the event that is triggered for a change in the state model for this tracker.	RO	Y	Integer

10.2 The object attributes *TaskType* and *PreviousTaskType* may take on one of the following values:

0. No Task — Value for *TaskType* when the EPT module is IDLE, and value for the *PreviousTaskType* when the EPT module transitions out of the IDLE state or initial state (no state).
1. Unspecified — A placeholder for unspecified task types.
2. Process — Tasks expected under normal or desired manufacturing operations that are capable of adding value by either (1) a physical-chemical change or (2) providing critical process information (e.g., inspection, metrology). This includes tasks that are dependent on a recipe ID, dependent on parameters (e.g., target etch thickness), or recipe and parameter independent (e.g., fixed purge time for all recipes or regardless target etch thickness).
3. Support — Tasks expected under normal or desired manufacturing operations that are incapable of adding value according to the criteria for a process task. These include alignment, handling and other transport, and environmental changes by supporting EPT modules (e.g., pumpdown and vent operations in a load lock).
4. Maintenance — Any task that is intended to change the state of the equipment for the purpose of maintaining equipment functionality or performance. This includes clean cycles, purges not expected under normal or desired manufacturing operations, and certain reset operations that do not put the EPT module or equipment in a non-operational state.
5. Diagnostics — Any task that obtains information about the status of the equipment for the purpose of determining the equipment's health or identifying an equipment problem(s). This excludes any task that is intended to change the status of the equipment, as in a maintenance task. This also excludes metrology and inspection tasks that provide critical information about the process even if those tasks also provide equipment diagnostic information as a byproduct.
6. Waiting — An equipment-initiated task that indicates the module is encountering normal waiting periods as part of its normal wafer processing scenario. A "Waiting" task applies when the module is awaiting a task to be completed by an external source before it can continue/complete processing (e.g., processing chamber waiting for robot to remove wafer, load port waiting for host ID/slot map verification, module waiting for a host command to initiate processing and commence material unloading.) Material must be present in the module when this task occurs.

10.2.1 NOTES: The following notes resolve ambiguities that arise from limitations of the equipment or EPT module point of view. From the equipment or EPT module point of view:

- Process maintenance activities (e.g., monitor wafers) and engineering activities (e.g., new process qualification runs) may be indistinguishable from normal or desired manufacturing operations. These operations shall be reported as either process or support operations.
- Equipment maintenance activities (e.g., equipment qualification runs monitor wafers) may be indistinguishable from normal or desired manufacturing operations. These operations shall be reported as either process or support operations.
- Activities executed for exception handling (e.g., pausing, aborting, stopping, or otherwise moving to a safe state) may be indistinguishable from support tasks. Examples include handling/transporting wafers to safe locations, pumping, venting, purging, etc. If the equipment or module implementing EPT does not have

reasonable inherent knowledge of purpose regarding these activities, the tasks should be always declared as support tasks. If the equipment or module implementing EPT does have reasonable inherent knowledge of purpose regarding these activities, these activities shall be either (1) modeled as task faults or otherwise blocked conditions, or (2) exempted entirely from treatment under EPT. In case 2, the affected EPT module or modules must be in a BLOCKED state due to related conditions reported to the EPT state model(s).

10.3 Blocked Reason Text Requirements

- Fault Condition: Blocked reason text should begin with “Fault: - ‘appropriate text’”
- Pause Condition: Blocked reason text should begin with “Pause: - ‘appropriate text’”
- Abort Condition: Blocked reason text should begin with “Abort: - ‘appropriate text’”

10.4 Configurable Disabling Of Event Reporting For Selected Transitions

10.4.1 The attribute DisableEventOnTransition may be used by the host to selectively disable EPT state model transition events for an individual EPTTracker object independently of whether the events are enabled at the equipment or not. This attribute is a simple list of integers 1–9 representing the disabled events.

10.4.2 For example, a list {1,4} indicates that no event reports should be sent for transitions 1 and 4. This attribute may be set by using the SEMI E39 service SetAttr for one or more objects.

11 EPT Events

11.1 This section defines the required events and variable data associated with transitions in the EPT state model.

11.2 Table 4 defines the data variables that are valid at the time of an EPT State Change event. When the equipment or EPT module triggers an EPTStateChange event, the value of the data variable shall be updated to reflect the data valid at the time the event occurred.

Table 4 Variable Data Definitions – Event Variables

VARIABLE NAME	DESCRIPTION	TYPE	ACCESS	COMMENT
EPTState	The new (resulting) EPT state of the module or equipment at the end of an EPT state transition.	Enumerated	RO	0 = IDLE 1 = BUSY 2 = BLOCKED
PreviousEPTState	The previous EPT state of the module or equipment of an end of an EPT state transition.	Enumerated	RO	0 = IDLE 1 = BUSY 2 = BLOCKED 3 = No State
EPTStateTime	Time spent in previous EPT state for the equipment or module.	Unsigned integer	RO	Seconds This is the time period between the entry into the previous EPT state and the entry into the current EPT State.
Clock	The time and date at the end of an EPT state transition.	Formatted numeric text: timestamp format	RO	When included in an event report, Clock represents the timestamp for the occurrence of the event.

<i>VARIABLE NAME</i>	<i>DESCRIPTION</i>	<i>TYPE</i>	<i>ACCESS</i>	<i>COMMENT</i>
TaskName	Name of the task that was started by the module or equipment at the occurrence of an EPT state transition.	Text, 0 to 80 characters	RO	TaskName is defined by the equipment supplier.
TaskType	The type of EPT Task currently running on this EPT module.	Enumerated	RO	0 = No Task 1 = Unspecified 2 = Process – adding value (e.g., exposing) 3 = Support – incapable of adding value (e.g., Handling/Transport) 4 = Equipment Maintenance (e.g., equipment initiated clean cycle) 5 = Equipment Diagnostics (e.g., equipment-initiated health check) 6 = Waiting (e.g., chamber waiting for a robot to remove a substrate)
PreviousTaskName	Name of the task that was completed by the module or equipment at the occurrence of an EPT state transition.	Text, 0 to 80 characters	RO	
PreviousTaskType	The type of the task that was completed by the module or equipment at the occurrence of an EPT state transition.	Enumerated	RO	0 = No Task 1 = Unspecified 2 = Process – adding value (e.g., exposing) 3 = Support – incapable of adding value (e.g., Handling/Transport) 4 = Equipment Maintenance (e.g., equipment initiated clean cycle) 5 = Equipment Diagnostics (e.g., equipment-initiated health check) 6 = Waiting (e.g., chamber waiting for a robot to remove a substrate)
BlockedReason	The number of the block condition that initiated the transition to the BLOCKED state for the module or equipment.	Enumerated	RO	0 – Not Blocked 1 – Unknown 2 – Safety Threshold 3 – Error Condition 4 – Parametric Exception 5 – Aborting, Aborted 6 – Pausing, Paused 7 – Reserved 8 – Reserved 9 – Reserved
BlockedReasonText	The reason why the transition was made to the BLOCKED state for the module or equipment.	Text, 0 to 80 characters	RO	Description of the Blocked reason

11.3 All events require the following data variables to be available:

< Clock > (Clock represents the timestamp for the occurrence of the event)
 < EPTState >
 <PreviousEPTState>
 < EPTStateTime >
 <TaskName>
 <TaskType>
 <PreviousTaskName>
 <PreviousTaskType>

Additional data required for specific transitions are shown in Table 5.

Table 5 Events

EVENT NAME	STATE MODEL TRANSITION NUMBER	DESCRIPTION	REQUIRED DATA
EPTStateChange ₀	1, 2, 3, 4, 5, 6, 7, 8, 9	Triggered by a EPT State change at the equipment-level.	<EqName > The following additional data is required for Transitions 5, 8 and 9: < Blocked Reason> < BlockedReasonText>
EPTStateChange _i	1, 2, 3, 4, 5, 6, 7, 8, 9	Triggered by an EPT State change at the EPT module-level. There exist i+1 EPTStateChange events, where i is the number of EPT modules. EPTStateChange ₁ is triggered by a state change at the 1 st EPT module, EPTStateChange ₂ is triggered by a state change at the 2 nd EPT module, etc.	<ModuleName _i > The following additional data is required for Transitions 5, 8 and 9: < Blocked Reason> < BlockedReasonText>

12 Requirements for Compliance

12.1 Table 6 provides a checklist for EPT compliance.

Table 6 EPT Compliance Statement

Fundamental EPT Requirements	EPT Section	Implemented	EPT Compliant
EPT State Model for Equipment	9	☐ Yes ☐ No	☐ Yes ☐ No
EPT State model for each EPT module	9	☐ Yes ☐ No	☐ Yes ☐ No
EPTTracker Objects	10	☐ Yes ☐ No	☐ Yes ☐ No
EPT Events	11	☐ Yes ☐ No	☐ Yes ☐ No

13 Related Documents

13.1.1 International SEMATECH Equipment Performance Management User Requirements Document

13.1.2 ISMT/J300E Equipment Performance Management Operator Requirements Document (URD)

13.1.3 Harel, D., “Statecharts: A Visual Formalism for Complex Systems,” Science of Computer Programming 8 (1987) 231-274.

RELATED INFORMATION 1

EXAMPLES OF EQUIPMENT PERFORMANCE TRACKING (EPT)

NOTICE: This related information is not an official part of SEMI E116 and was derived from North American Information and Control. This related information was approved for publication by full letter ballot procedures on April 30, 2002.

R1-1 Examples of EPT Modules per Tool Type

R1-1.1 Table R1-1 lists examples of EPT modules for various equipment types. It is not required that each tool type support all or be limited to only the modules listed; rather, the listing is provided as an example only.

Table R1-1 Example EPT Modules

<i>Tool Type</i>	<i>Modules</i>								
<i>CMP/Planar</i>	Loadports	Alignment Stage	Wafer Handler	Transport	Polish Table	Scrub Station	Rinse Station		
<i>Linked Litho Track</i>	Loadports	Central Wafer Handler Module	Spin Cup	Link Transfer Station	Buffer	Vapor Prime	Chill Plate	Soft Bake Oven	
<i>Linked Litho Expose</i>	Expose Stage	Wafer Transfer	Link Transfer Station – In port	Link Transfer Station – Out port	Reticle Stage	Reticle Library	Reticle Loader Arm		
<i>Thin Film Sputter</i>	Loadports	Central Wafer Handler(s)	Cool Down	PVD	Airlock	Argon Sputter Etch	Degas	In Situ Metrology	CVD
<i>Diffusion Furnace</i>	Loadports	Quartz Loading Station	Reactor Module	Wafer Transfer System	Buffer				
<i>Implanter</i>	Loadports	Central Wafer Handler(s)	Implant Chamber	Platen	Notch Aligner	Coldplate			
<i>Plater</i>	Loadports	Central Wafer Handler	Degas	Plate Chamber					
<i>Thermal Anneal</i>	Loadports	Central Wafer Handler	Bake Stage						
<i>Wet Etcher</i>	Loadports	Central Wafer Handler	Aligner	Bath CH 1	Bath CH 2	Buffer			
<i>Dry Etcher</i>	Loadports	Central Wafer Handler	Aligner	Chill Plate	Etch CH 1	Etch CH 2			
<i>Wafer Sorter</i>	Loadports	Central Wafer Handler	Aligner						

<i>Tool Type</i>	<i>Modules</i>								
<i>Metrology</i>	Loadports	Central Wafer Handler	Align-ment Stage	Measure-ment Stage					
<i>Inspection</i>	Loadports	Inspection Stage	Handler						
<i>Probe / Test</i>	Loadports	Probe/ Test Platform	Handler						

R1-2 EPT Module / Task Definition Example

R1-2.1 Figure R1-1 shows a simple module and the associated actions/tasks that correspond. The EPT Tasks have been diluted to the level that will allow a serial chain of tasks to occur. The Internal Tasks column is present to show that the equipment may have much more happening concurrently at a more detailed level. Table R1-2 shows a possible task-based process flow in this module.

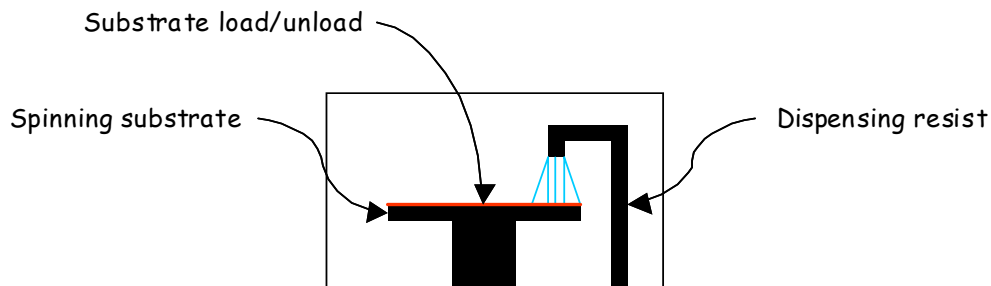


Figure R1-1
Example EPT Module

Table R1-2 Example EPT Tasks

<i>Step</i>	<i>Action</i>	<i>Possible Internal Tasks</i>	<i>EPT Active Task</i>
1	Loading the substrate	Receiving the substrate Substrate alignment	Substrate load
2	Coating the substrate	Substrate spin (accelerate) Substrate spin (steady) Dispense resist Substrate spin (decelerate)	Substrate coat
3	Unloading the substrate	Substrate alignment Substrate removal	Substrate unload

R1-3 An Example of Equipment Performance Tracking on a Carrier Handling Module

R1-3.1 This section provides examples of equipment performance tracking to clarify the application of the standard to modules that are part of the Equipment Front-End Module (EFEM).

R1-4 Carrier Handling Module (CHM)

R1-4.1 The Carrier Handler is a part of EFEM, excluding any Substrate Handler, which is defined in SEMI E101. A Carrier Handling Module is similar but different from the definition. It is a more general subsystem to handle carriers without handling substrates in EFEM that is applicable to various analyses and applications. The Carrier Handling Module is a logical module that may or may not represent an existing physical module.

R1-4.2 The advantages of applying Equipment Performance Tracking to the Carrier Handling Module are as follows:

- Any equipment, except certain specific ones such as an expose tool within a linked litho tool in the lithography process, has one or more Carrier Handling Modules.

The state of the Carrier Handling Module leads to whole input and output of material to and from an equipment.

- Monitoring the Equipment Performance Tracking state of the Carrier Handling Module helps control loading to or unloading from equipment on appropriate time.
- The state can reveal what related SEMI standards, i.e. SEMI E87 and SEMI E90, may not address.
- Estimation of capabilities about sub-components on a Carrier Handling Module is available with tracking the state.
- Each component that makes up the Carrier Handling Module may experience mechanical failure. This can cause the entire Carrier Handling Module to be in an out of service state. Tracking time between failures would be valuable.

R1-5 Components to track performance of Carrier Handling Module

R1-5.1 Generally, a Carrier Handling Module has the following components.

- Carrier Handoff PIO (SEMI E84) Port
- Carrier Transfer Mechanism
- Carrier ID Reader
- Carrier ID Writer
- FOUP Opener
- Carrier Slot Mapper

R1-5.2 This section investigates the following composition to represent examples for Fixed Buffer type and Internal Buffer type equipment. An example of a Carrier Handling Module is illustrated in Figure R1-2. Number of installed components are listed in the Table R1-3.

Table R1-3 Typical Composition of Carrier Handoff Module

<i>Number Installed</i>	<i>Carrier Handoff PIO</i>	<i>Carrier Transfer Mechanism</i>	<i>Carrier ID Reader</i>	<i>Carrier ID Writer</i>	<i>FOUP Opener</i>	<i>Slot Mapper</i>
Internal Buffer Type	2	4	2	1	1	1
Fixed Buffer Type	2	2	2	2	2	2

^{#1} Several Carrier Transfer Mechanisms are usually docking mechanism for Fixed Buffer type. For Internal Buffer type, one of them is a Carrier Transfer Robot in internal buffer, one is a Docking Mechanism inside the internal buffer and the other one is Carrier Transfer Mechanism to pass carrier between Load Port and the Internal Buffer.

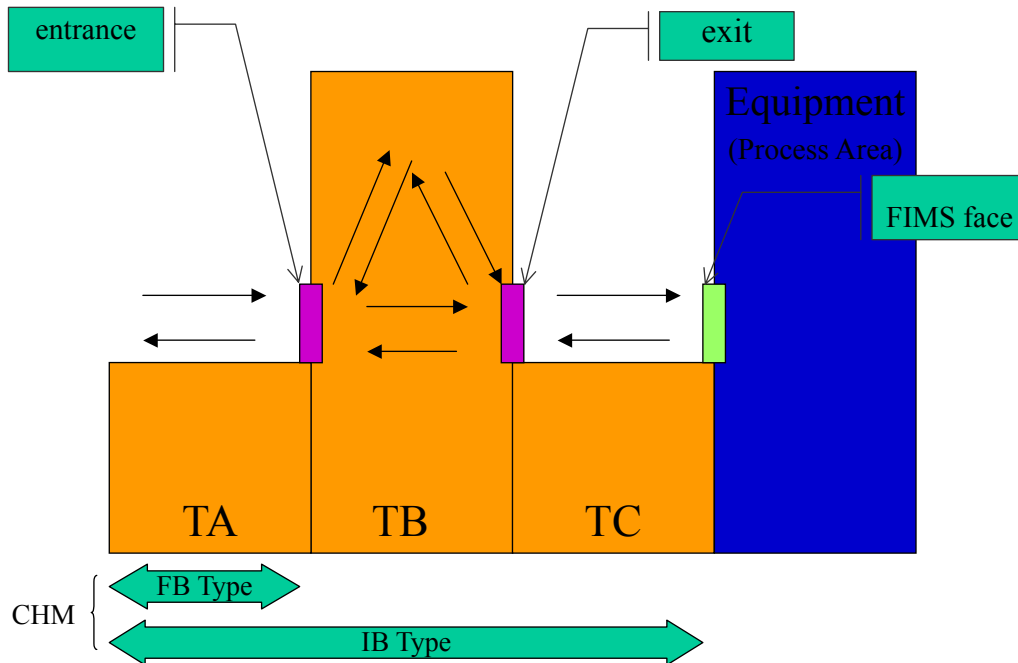


Figure R1-2
Carrier Handling Module

In Diagram R1-2, the following translation applies

FB Type = Fixed Buffer Type

IB Type = Internal Buffer Type

TA = Transport A

TB = Transport B

TC = Transport C

R1-6 Triggers of Transition for each Components of Carrier Handling Module

The following tables describe triggers to change the state of components for specific tasks.

Table R1-4 Triggers of Transitions for Specific Tasks of Carrier Handoff PIO

#	Previous State	Trigger	New State	Action(s)	Comment
2	IDLE	Receiving VALID signal in SEMI E84.	BUSY	Set off handoff sequence	Loading or Unloading sequence. Processing or moving wafers between carrier and process container before Unloading.
3	BUSY	Receiving COMPT signal in SEMI E84 or Timeout in handoff sequence.	IDLE	Waiting disconnected	Loading or Unloading sequence is completed or terminated. Processing or moving wafers between carrier and process container after Loading.

Table R1-5 Triggers of Transitions for Specific Tasks of Carrier Transfer Mechanism on Loadport of Fixed Buffer Type or FIMS Port of IB

#	Previous State	Trigger	New State	Action(s)	Comment
2	IDLE	Docking or Undocking Started	BUSY	None	Carrier moving sequence before docking. Processing or moving wafers between carrier and process container before undocking.
3	BUSY	Docking or Undocking Complete	IDLE	None	Processing or moving wafers between carrier and process container after docking.

Table R1-6 Triggers of Transitions for Specific Tasks of Carrier Transfer Mechanism in Internal Buffer Unit

#	Previous State	Trigger	New State	Action(s)	Comment
2	IDLE	Transfer Started	BUSY	Moves carrier to destination	
3	BUSY	Transfer Complete	IDLE	None	Ready to unload or remove from port.

Table R1-7 Triggers of Transitions for specific tasks of Carrier ID Reader

#	Previous State	Trigger	New State	Action(s)	Comment
2	IDLE	Reading Started	BUSY	None	Carrier moving sequence before reading.
3	BUSY	Reading Complete	IDLE	Verification	Processing or moving wafers between carrier and process container after opening.

Table R1-8 Triggers of Transitions for Specific Tasks of Carrier ID Writer

#	Previous State	Trigger	New State	Action(s)	Comment
2	IDLE	Writing Started	BUSY	None	Processing wafers before writing.
3	BUSY	Writing Complete	IDLE	Waiting next writing or any post writing action	Carrier moving sequence after writing.

Table R1-9 Triggers of Transitions for Specific Tasks of FOUP Opener

#	Previous State	Trigger	New State	Action(s)	Comment
2	IDLE	Opening or Closing Started	BUSY	None	Carrier moving sequence before opening.
3	BUSY	Opening or Closing Complete	IDLE	None	Processing or moving wafers between carrier and process container after opening.

Table R1-10 Triggers of Transitions for Specific Tasks of Slot Mapper

#	Previous State	Trigger	New State	Action(s)	Comment
2	IDLE	Scanning Started	BUSY	None	Carrier moving sequence before mapping.
3	BUSY	Scanning Complete	IDLE	Waiting next scan	Processing or moving wafers between carrier and process container after mapping.

Table R1-11 Triggers of Transitions Common for all Components of Carrier Handling Module

#	Previous State	Trigger	New State	Action(s)	Comment
1	(no state)	Initialization is complete	IDLE	None	
7	BLOCKED	Get Recovered	IDLE	None	
8	IDLE	Unable to do task	BLOCKED	None	
9	BLOCKED	Get another Fault	BLOCKED	None	

R1-7 Operation Scenarios for Carrier Handling Module

R1-7.1 A couple of following scenarios are typical for different type of Carrier Handling Module.

R1-7.2 Carrier Handling Module for Fixed Buffer Type

R1-7.2.1 Beside the number of Loadports, following table shows just one set of components out of them.

R1-7.2.2 Indicated subsystems: PIO = Carrier Handoff PIO Port, CID-R = Carrier ID Reader, X'fer Transfer Mechanism = Carrier Transfer Mechanism (Docking Mechanism), CID-W = Carrier ID Writer, Opener = FOUP Opener, Mapper = Carrier Slot Mapper.

Table R1-12 Scenario of Fixed Type Carrier Handling Module

Event #	Description	Time	PIO	CID-R.	X'fer Mech	CID-W	Opener	Mapper
0	Carrier Handling Module and all components starting in "IDLE"	0:00	IDLE	IDLE	IDLE	IDLE	IDLE	IDLE
1	Loading Started	0:30	BUSY	↑	↑	↑	↑	↑
2	Loading Completed	1:00	IDLE	↑	↑	↑	↑	↑
3	ID Reading Started	1:05	↑	BUSY	↑	↑	↑	↑
4	ID Reading Complete	1:10	↑	IDLE	↑	↑	↑	↑
5	ID Verification	-	↑	↑	↑	↑	↑	↑
6	Docking Started	1:20	↑	↑	BUSY	↑	↑	↑
7	Docking Complete	1:50	↑	↑	IDLE	↑	↑	↑
8	Opening Start	1:55	↑	↑	↑	↑	BUSY	↑
9	Opening Complete	2:25	↑	↑	↑	↑	IDLE	↑
10	Mapping Started	2:30	↑	↑	↑	↑	↑	BUSY
11	Mapping Complete	2:15	↑	↑	↑	↑	↑	IDLE
12	Slotmap Verification	-	↑	↑	↑	↑	↑	↑
13	Process Started	12:15	↑	↑	↑	↑	↑	↑
14	Process Completed	47:15	↑	↑	↑	↑	↑	↑
15	Writing Started	47:20	↑	↑	↑	BUSY	↑	↑
16	Writing Complete	48:10	↑	↑	↑	IDLE	↑	↑
17	Closing Start	48:10	↑	↑	↑	↑	BUSY	↑
18	Closing Complete	48:40	↑	↑	↑	↑	IDLE	↑
19	Undocking Started	48:45	↑	↑	BUSY	↑	↑	↑
20	Undocking Complete	49:15	↑	↑	IDLE	↑	↑	↑
21	Unloading Started	52:25	BUSY	↑	↑	↑	↑	↑
22	Unloading Complete	52:55	IDLE	↑	↑	↑	↑	↑
23	Carrier Handling Module and all components returning in "IDLE"	52:55	↑	↑	↑	↑	↑	↑

R1-7.3 Carrier Handling Module for Internal Buffer Type

R1-7.3.1 It is assumed that Carrier Slot Mapper and Carrier ID Writer are mounted at an FIMS port next to Internal Buffer located inside of the equipment. The following table shows the set of components of a single Loadport.

R1-7.3.2 Indicated subsystems: PIO = Carrier Handoff PIO Port, CID-R = Carrier ID Reader, X'fer A = Carrier Transfer Mechanism from Handoff position to Carrier In/Out position, X'fer B = Carrier Transfer Mechanism of Internal Buffer, X'fer C = Carrier Transfer Mechanism (Docking Mechanism) for internal FIMS port, CID-W = Carrier ID Writer, Opener = FOUP Opener, Map = Carrier Slot Mapper

Table R1-13 Scenario of Fixed Type Carrier Handling Module

Event #	Description	Time	PIO	CID-R.	X'fer A	X'fer B	X'fer C	CID-W	Opener	Map
0	Carrier Handling Module and all components starting in "IDLE"	0:00	IDLE	IDLE	IDLE	IDLE	IDLE	IDLE	IDLE	IDLE
1	Loading Started	0:30	BUSY	↑	↑	↑	↑	↑	↑	↑
2	Loading Completed	1:00	IDLE	↑	↑	↑	↑	↑	↑	↑
3	ID Reading Started	1:05	↑	BUSY	↑	↑	↑	↑	↑	↑
4	ID Reading Complete	1:10	↑	IDLE	↑	↑	↑	↑	↑	↑
5	ID Verification	-	↑	↑	↑	↑	↑	↑	↑	↑
6	Transfer Started	1:20	↑	↑	BUSY	↑	↑	↑	↑	↑
7	Transfer Complete	1:35	↑	↑	IDLE	↑	↑	↑	↑	↑
8	Moving Started	1:40	↑	↑	↑	BUSY	↑	↑	↑	↑
9	Moving Complete	2:25	↑	↑	↑	IDLE	↑	↑	↑	↑
10	Docking Started	2:30	↑	↑	↑	↑	BUSY	↑	↑	↑
11	Docking Complete	2:50	↑	↑	↑	↑	IDLE	↑	↑	↑
12	Opening Start	2:55	↑	↑	↑	↑	↑	↑	BUSY	↑
13	Opening Complete	3:25	↑	↑	↑	↑	↑	↑	IDLE	↑
14	Mapping Started	3:30	↑	↑	↑	↑	↑	↑	↑	BUSY
15	Mapping Complete	3:15	↑	↑	↑	↑	↑	↑	↑	IDLE
16	Slotmap Verification	-	↑	↑	↑	↑	↑	↑	↑	↑
17	Wafer Moving Started	3:30	↑	↑	↑	↑	↑	↑	↑	↑
18	Wafer Moving Completed	7:00	↑	↑	↑	↑	↑	↑	↑	↑
12	Closing Start	7:05	↑	↑	↑	↑	↑	↑	BUSY	↑
13	Closing Complete	7:20	↑	↑	↑	↑	↑	↑	IDLE	↑
21	Undocking Started	7:25	↑	↑	↑	↑	BUSY	↑	↑	↑
22	Undocking Complete	7:45	↑	↑	↑	↑	IDLE	↑	↑	↑
8	Moving Started	7:50	↑	↑	↑	BUSY	↑	↑	↑	↑
9	Moving Complete	8:35	↑	↑	↑	IDLE	↑	↑	↑	↑
17	Process Started	16:15	↑	↑	↑	↑	↑	↑	↑	↑
18	Process Completed	147:15	↑	↑	↑	↑	↑	↑	↑	↑
8	Moving Started	146:30	↑	↑	↑	BUSY	↑	↑	↑	↑
9	Moving Complete	147:15	↑	↑	↑	IDLE	↑	↑	↑	↑
10	Docking Started	147:20	↑	↑	↑	↑	BUSY	↑	↑	↑
11	Docking Complete	147:40	↑	↑	↑	↑	IDLE	↑	↑	↑
19	Writing Started	147:45	↑	↑	↑	↑	↑	BUSY	↑	↑
20	Writing Complete	148:10	↑	↑	↑	↑	↑	IDLE	↑	↑
12	Opening Start	148:10	↑	↑	↑	↑	↑	↑	BUSY	↑
13	Opening Complete	148:25	↑	↑	↑	↑	↑	↑	IDLE	↑
17	Wafer Moving Started	148:30	↑	↑	↑	↑	↑	↑	↑	↑
18	Wafer Moving Completed	152:00	↑	↑	↑	↑	↑	↑	↑	↑
12	Closing Start	152:05	↑	↑	↑	↑	↑	↑	BUSY	↑
13	Closing Complete	152:20	↑	↑	↑	↑	↑	↑	IDLE	↑
21	Undocking Started	152:25	↑	↑	↑	↑	BUSY	↑	↑	↑
22	Undocking Complete	152:45	↑	↑	↑	↑	IDLE	↑	↑	↑
23	Moving Started	152:50	↑	↑	↑	BUSY	↑	↑	↑	↑
24	Moving Complete	153:35	↑	↑	↑	IDLE	↑	↑	↑	↑
25	Transferring Started	153:40	↑	↑	BUSY	↑	↑	↑	↑	↑
26	Transferring Complete	153:55	↑	↑	IDLE	↑	↑	↑	↑	↑



27	Unloading Started	202:25	BUSY	↑	↑	↑	↑	↑	↑	↑
28	Unloading Complete	202:55	IDLE	↑	↑	↑	↑	↑	↑	↑
29	Carrier Handling Module and all components returning in "IDLE"	202:55	↑	↑	↑	↑	↑	↑	↑	↑

R1-8 Tracking Performance for Whole Carrier Handling Module

R1-8.1 Even composition and property of equipment are dependent on equipment type, operation and usage depends on equipment user. Because actual usage deeply affects the whole performance of the Carrier Handling Module and/or Equipment, the user is responsible for reconstructing the performance from performance tracking events for each component.

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SEMI E116.1-1104

SPECIFICATION FOR SECS-II PROTOCOL FOR EQUIPMENT PERFORMANCE TRACKING (EPT)

This specification was technically approved by the Global Information & Control Committee and is the direct responsibility of the North American Information & Control Committee. Current edition approved by the North American Regional Standards Committee on August 16, 2004. Initially available at www.semi.org September 2004; to be published November 2004. Originally published March 2003.

NOTICE: This document was rewritten in its entirety in 2004.

1 Purpose

1.1 This document maps the services and data of the Equipment Performance Tracking Standard (EPT) to SECS-II streams, functions, and data definitions.

2 Scope

2.1 This document applies to all implementations of EPT that use the SECS-II message protocol (SEMI E5). Compliance to this standard requires compliance to both SEMI E116 and SEMI E5.

NOTICE: This standard does not purport to address safety issues, if any, associated with its use. It is the responsibility of the users of this standard to establish appropriate safety and health practices and determine the applicability of regulatory or other limitations prior to use.

3 Referenced Standards

3.1 SEMI Standards

SEMI E5 — SEMI Equipment Communications Standard 2 Message Content (SECS-II)

SEMI E30 — Generic Model for Communications and Control of Manufacturing Equipment (GEM)

SEMI E39 — SECS-II Protocol for Object Services Standard (OSS)

SEMI E116 — Provisional Specification for Equipment Performance Tracking (EPT)

NOTICE: Unless otherwise indicated, all documents cited shall be the latest published versions.

4 SECS-II Attribute Definitions

4.1 Table 1 provides the SECS-II structure definitions for the E116 EPTTracker Object. This must be compliant with the E39 (OSS) protocol. The host does not have the ability to create or destroy these objects. However, the host must be able to access all attributes via the default E39 GetAttr services (for all attributes) and SetAtt services (for RW attributes).

4.2 A subset of these object attributes are enumerated via SECS-II Data Variables that may be provided via event-reports. See the section on Variable Data Item Mapping in this document.

Table 1 EPTTracker Object SECS-II Attributes Definitions

<i>Attribute Name</i>	<i>Attribute Data Form: SECS-II Structure</i>
“ObjType”	20 (A[10]) must always be set to “EPTTracker”
“ObjID”	20 (A[1..80]) <OBJECTIDENTIFIER> ObjID is equipment defined. Examples are “EQUIPMENT”, “CHAMBER A”, “ROBOT ARM”. Conforms to the restrictions of ObjID as specified in SEMI E39.1
“BlockedReason”	51 (U1) BlockedReason where the values are enumerated as: 0 = Not Blocked 1 = Unknown 2 = Safety Threshold 3 = Error Condition 4 = Parametric Exception 5 = Aborting, Aborted 6 = Pausing, Paused 7 = Reserved 8 = Reserved 9 = Reserved
“BlockedReasonText”	20 (A) BlockedReasonText Usage is equipment defined. Examples: “Abort – Host-requested Abort”, “Fault – Reticle Stage Error”; “Pause – Host-requested Pause”
“EPTElementType”	51 (U1) EPTElementType where the values are enumerated as: 0 = Equipment 1 = Production EPT Module 2 = EFEM/Load Port EPT Module
“EPTState”	51 (U1) EPTState where the values are enumerated as: 0 = Idle 1 = Busy 2 = Blocked
“EPTStateTime”	54 (U4) EPTStateTime
“EPTElementName”	20 (A) EPTElementName Usage is equipment defined. Examples “EQUIPMENT”, “CHAMBER A”, “ROBOT ARM”.
“PreviousEPTState”	51 (U1) PreviousEPTState where the values are enumerated as: 0 = Idle 1 = Busy 2 = Blocked 3 = No State
“PreviousTaskName”	20 (A[0..80]) PreviousTaskName Usage is equipment defined. Examples: “Loading”, “Etching”, “Pumping Down”

“PreviousTaskType”	51 (U1) PreviousTaskType where the values are enumerated as: 0 = No Task 1 = Unspecified 2 = Process -- adding value (e.g., exposing) 3 = Support -- incapable of adding value (e.g., Handling/Transport) 4 = Equipment Maintenance (e.g., Equipment-initiated clean cycle) 5 = Equipment Diagnostics (e.g., Equipment-initiated health check) 6 = Waiting (e.g., chamber waiting for a robot to remove a substrate)
“TaskName”	20 (A[0..80]) TaskName Usage is equipment defined. Examples: “Loading”, “Etching”, “Pumping Down”
“TaskType”	51 (U1) TaskType where the values are enumerated as: 0 = No Task 1 = Unspecified 2 = Process -- adding value (e.g., exposing) 3 = Support -- incapable of adding value (e.g., Handling/Transport) 4 = Equipment Maintenance (e.g., Equipment-initiated clean cycle) 5 = Equipment Diagnostics (e.g., Equipment-initiated health check) 6 = Waiting) e.g., chamber waiting for a robot to remove a substrate)
“TransitionTimeStamp”	20 (A[16]) YYYYMMDDhhmmsscc YYYY=Year 0000 to 9999 MM=Month 01 to 12 DD=Day 01 to 31 hh=Hour 00 to 23 mm=Minute 00 to 59 ss=Second 00 to 59 cc= Centisecond 00 to 99
“Transition”	51 (U1) Transition where the values are enumerated as: 1 = Transition 1 2 = Transition 2 . . . 9 = Transition 9
“TrackerEventID”	54 (U4) TrackerEventID where: This is identical to EPTStateChangeEvent_i corresponding to this object (see the Events Mapping section in this document)
“DisableEventOnTransition”	L,n 1. 51 (U1) <Transition Number> 2. 51 (U1) <Transition Number> ... n. 51 (U1) <Transition Number> where the elements of this list are can only have values in the range 1..9, in order, with no repeats. This list may be empty.



5 Events Mapping

5.1 Table 2 shows the specific SECS-II streams and functions that shall be used for SECS-II implementations of the events defined in EPT. A single collection event is generated for each of the equipment and the modules to report transitions occurring within the state model. This approach differs from state models that are defined in other 300 mm-related SEMI standards, which utilize a unique CEID for each transition in the state model. For example, a piece of equipment with 3 EPT modules would have 4 associated CEIDs. The CEID indexed 0 would report equipment-level events and 3 CEIDs corresponding to the equipment's 3 EPT modules indexed 1...3 would report EPT module-level events.

Table 2 Event Mapping Table

<i>Event Name</i>	<i>Stream, Function</i>	<i>SECS-II Message Name</i>
EPTStateChangeEvent _i	S6F11/F12	Event Report Send/Acknowledge

6 Variable Data Item Mapping

6.1 Table 3 shows the specific SECS-II data classes, and formats needed for SECS-II implementations of SEMI E116 EPT variable data items. The data variables must be updated as specified in the E116 standard when a transition occurs in the state-model for any of the EPTTracker objects implemented on the equipment. The Variable Names below match the object Attribute Names from which the variables are to be updated (see the section on EPT Tracker Object SECS-II Attribute Definitions)

These variables are tied to the occurrence of transitions in the state model and are not impacted by the DisableEventOnTransition attribute in the EPTTracker object.

Table 3 Event Variable Data Item Mapping Table

<i>Variable Name</i>	<i>Class</i>	<i>Format</i>
EPTState	DVVAL	51 (U1)
PreviousEPTState	DVVAL	51 (U1)
EPTStateTime	DVVAL	54 (U4)
TaskName	DVVAL	20 (A[0..80])
TaskType	DVVAL	51 (U1)
PreviousTaskName	DVVAL	20 (A[0..80])
PreviousTaskType	DVVAL	51 (U1)
BlockedReason	DVVAL	51 (U1)
BlockedReasonText	DVVAL	20 (A)

Table 4 Additional Data Item Mapping Table

<i>Variable Name</i>	<i>SECS-II Data Item</i>
Clock	CLOCK

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